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A new lipophilic amino alcohol, chemically similar to compound FTY720, attenuates the pathogenesis of experimental autoimmune encephalomyelitis by PI3K/Akt pathway inhibition

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ABSTRACT

Experimental autoimmune encephalomyelitis (EAE) is one of the main animal models used for the study of Multiple Sclerosis (MS). Long-chain lipophilic amino alcohols with immunoregulatory activities have already been studied in some models of inflammatory diseases, but the action of these compounds in EAE and MS is still unknown. In this study, we investigated whether the lipophilic amino alcohol 4b would act to improve the clinical signs of EAE and reduce the demyelination process and the neuroinflammatory parameters in the spinal cord, as well as the inflammatory process in the inguinal lymph nodes, of C57Bl/6 mice induced with EAE after stimulation with MOG₃₅₋₅₅ and pertussis toxin. The 4b treatment (1.0 mg/kg/day) was orally administered, starting on the day of onset of clinical signs of the disease (10th) and ending on the 20th day after immunization. This treatment was able to reduce the cell count on the inguinal lymph nodes, the migration of inflammatory cells into the central nervous system (CNS), as well as the processes of microgliosis, astrogliosis, and the production of chemokines and pro-inflammatory cytokines, thus increasing the IL-10 anti-inflammatory cytokine levels in EAE mice. The inhibition of Akt phosphorylation in the CNS of EAE mice after treatment with 4b indicates that the immunoregulatory action of 4b is related to the PI3K/Akt signaling pathway. Our results indicate the immunoregulatory efficacy of the new compound 4b in the control of some inflammatory parameters and in the glial proliferation. In addition, 4b was able to reduce the demyelination of neurons and the worsening of clinical signs of EAE as effectively as the compound FTY720, the first oral drug approved by the FDA for the treatment of MS.

1. Introduction

Multiple sclerosis (MS) is a chronic inflammatory disease with autoimmune origin characterized by the demyelination of axons of neurons in the central nervous system (CNS). It can provoke neurological damages, sensory changes, and motor disabilities, causing pain, fatigue, and even the paralysis of limbs in patients [1]. Currently, more than 2.5 million individuals are affected by MS, generating an estimated cost of US\$ 2–3 billion per year in public health and inflicting psychological problems on patients since it affects their quality of life and families [2]. MS has a complex etiology; however, the influx of autoreactive

immune cells in the CNS is one of the immunological factors involved in this disease [3]. Studies have shown that the inflammatory response is the hallmark of all forms of the disease, leading to the activation of CNS-resident cells, such as microglia and astrocytes. Activated microglia are common in demyelination plaques and in the normal-appearing extraleSIONAL white matter in the CNS of patients diagnosed with MS [4–6]. Microglia are gliocytes capable of phagocytosing and processing antigens in the CNS [7] that can be exaggeratedly activated, thus producing a variety of molecules that are harmful to neurons and oligodendrocytes, including free radicals and pro-inflammatory cytokines [8]. If the activation of microglia and astrocytes is persistent, these cell

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products can become toxic and worsen the process of progressive axonal degeneration in MS and in some of its animal models, such as the Experimental Autoimmune Encephalomyelitis (EAE) [5,9,10]. The traditional therapies for MS are still not diverse and do not benefit patients with progressive forms of the disease. Therefore, it is still necessary to investigate new therapeutic strategies for the treatment of MS, which are safe, effective, but mainly low-cost, since most of the therapies approved today for the control of the disease have a high cost for patients and health care services [11]. The phosphatidylinositol 3-kinase (PI3K)/ protein kinase B (PKB/Akt)/ mechanistic target of rapamycin (mTOR) signaling pathway has been linked to the pathogenesis of EAE since it acts in important biological processes, such as survival, proliferation, and migration of immune cells. A previous study showed that splenocytes derived from EAE animals presented an increase in the activation of this signaling pathway, which was inhibited by treatment with the compound FTY720, the first oral drug approved by the FDA for the treatment of MS [12]. In previous studies, we have investigated the role of long-chain lipophilic amino alcohols in reducing inflammatory mediators in carrageenan-induced mice paw edema and delayed-type hypersensitivity mice paw edema models, by their immunosuppressive activities [13]. Thus, the present study aimed to investigate if the compound 4b – a new lipophilic amino alcohol structurally similar to FTY720 – would reduce the clinical signs of MS and attenuate the demyelination process, and also if 4b would improve the regulation of neuroinflammatory parameters that occur after EAE induction in mice.

2. Materials and methods

2.1. EAE induction and clinical evaluations

Female C57BL/6 mice (n = 68; 6–8 weeks old) were obtained from the Animal Care Facilities of the Federal University of Juiz de Fora (UFJF), and they were maintained in microisolator cages in the animal facility of the Center for Cellular Technology and Applied Immunology (IMUNOCET). The protocol used for the EAE induction was approved by the Brazilian Institutional Ethics Committee (CEUA) of UFJF (Protocol number 019/2018). We are concerned with avoiding discomfort, such as stress, in animals. Initially, each animal was immunized subcutaneously (sc) on each side of the dorsal region near the base of the tail with 100 µg of the myelin oligodendrocyte glycoprotein peptide (MOG₃₅₋₅₅; GenOne Biotechnologies, Rio de Janeiro, Brazil) emulsified in Freund's Complete Adjuvant (Sigma- Aldrich, Saint Louis, USA) in the proportion of 1:1, supplemented with 400 µg of *Mycobacterium tuberculosis* (H37RA; Difco, Detroit, USA). On the day of the immunization and after 48 h, each animal received 300 ng of pertussis toxin from *Bordetella pertussis* (Sigma-Aldrich, Saint Louis, USA) intraperitoneally (ip). The animals (n = 17/group) were evaluated individually for neurological disability generated by EAE, according to a previous study by our group [14]. The final clinical score was obtained by the arithmetic mean of the score values for each group calculated on each day of analysis.

2.2. Drug treatments

The 1-aminotetradecan-2-ol, a lipophilic amino alcohol compound herein called 4b, was synthesized and kindly provided by Professor Mauro V. Almeida, from the Chemistry Department of UFJF. Its synthesis started with the mesylation of alcohol 1b in pyridine at 0 °C, giving rise to intermediate compound 2b. Mesylate compound 2b was treated with sodium azide in DMF (dimethylformamide) at 120 °C, giving rise to a monoazide 3b, which was subsequently hydrogenated, resulting in the amino alcohol 4b. The compound was purified and characterized by infrared spectroscopy and hydrogen and carbon nuclear magnetic resonance (1H NMR and 13C NMR, respectively) (Fig. 1) [13]. The 4b compound showed significant results in reducing clinical signs in animal models of carrageenan-induced paw edema and

ovalbumin-induced late hypersensitivity in animals of the BALB/c strain [13], but its effects on the EAE model are unknown. FTY720 (Sigma Chemical Co, Saint Louis, USA) was dissolved in filtered water and orally administered (1.0 mg/kg/day) when the initial symptoms of EAE started, on the 10th day post-immunization (dpi). 4b was diluted in filtered water with 5% DMSO and administered by gavage (1.0 mg/kg/day), from the 10th dpi. The mice were randomly divided into four groups: (1) Control group (the animals received only vehicle); (2) EAE + vehicle; (3) EAE + FTY720; (4) EAE + 4b.

2.3. Hematoxylin and eosin (H&E) and Luxol Fast Blue (LFB) staining

To perform the histological analysis of the spinal cord, on the 21st dpi, the mice (n = 5/group) were anesthetized with ketamine and xylazine in the proportion of 80:10 mg/kg, respectively, and then perfused with phosphate-buffered saline followed by paraformaldehyde 4%, transcardially. From each collected spinal cord, the thoracic and lumbar regions were sectioned, subsequently fixed in 4% paraformaldehyde, and embedded in paraffin to generate slides. The sections (5 µm) were stained with hematoxylin and eosin (Sigma-Aldrich, Saint Louis, USA) to assess tissue damage and infiltration of inflammatory cells. The sections in the slides were coverslipped with the mounting medium Entellan® (Merck KGaA, Darmstadt, Germany). The images were captured by an Olympus BX53 microscope at ×20 magnification. Semi-quantitative histological evaluation for inflammation was analyzed blindly, using a scoring scale described by Xiao and cols. (2015) [15]: 0, without inflammation; 1. cellular immune infiltration only in the perivascular area and meninges; 2. mild cell infiltration into the parenchyma; 3. moderate cell infiltration in the parenchyma; 4. severe cell infiltration in the parenchyma [15,17]. Moreover, to assess the demyelination, 10 µm spinal cord sections were stained with Luxol Fast Blue (LFB)- cresyl violet (Sigma-Aldrich, Saint Louis, USA). The sections in the slides were coverslipped with the mounting medium Entellan® (Merck KGaA, Darmstadt, Germany). The images were captured by an Olympus BX53 microscope at ×20 magnification. Demyelination foci were assessed using a scoring scale described by Abdul-Majid and cols (2003) and Dias and cols (2018) [16,17]: +/– traces of perivascular or subpial demyelination (score 1); + marked perivascular or subpial demyelination (score 2); ++ confluent or subpial perivascular demyelination (score 4); +++ massive confluent demyelination (for example, approximately half of the spinal cords in foci of demyelination) (score 6); ++++ extensive demyelination (complete demyelination) (score 8).

2.4. Immunofluorescence staining

For the procedures for the detection of microglia (Iba-1 + cells) and astrocytes (GFAP + cells), spinal cord sections of 5 µm (n = 5/group) were dewaxed with xylene and subsequently hydrated in an ethanol gradient (100, 95, 90, 80 and 70%, sequentially). For the antigenic recovery of the samples, the tissue sections were then incubated for 30 min with citrate buffer at 70 °C. Non-specific connections were blocked by using the blocking solution [5% bovine serum albumin (BSA) and 0.2% Tween 20 in PBS] for 1 h. Then, the tissue sections were incubated at 4 °C overnight with the respective primary antibodies: rabbit anti-Iba-1 (1:500; Wako Chemicals, Osaka, Japan) and mouse anti-GFAP (1:500; Cell Signaling, Danvers, MA, USA). The tissue sections were incubated at 4 °C overnight with the respective secondary antibodies: Alexa Fluor 594 anti-rabbit (1: 1000; Invitrogen, Carlsbad, CA, USA) and Alexa Fluor 488 anti-mouse (1: 1000; Invitrogen, Carlsbad, CA, USA). After this incubation period, the sections were washed, and the slides were mounted using the Fluoromount mounting medium (Sigma-Aldrich, St. Louis, MO, USA). The sections were photographed with a digital camera coupled to an Olympus fluorescence microscope, and the images were analyzed using the MacBiophotonics ImageJ software (NIH, Bethesda MD, USA). For the analysis, we used

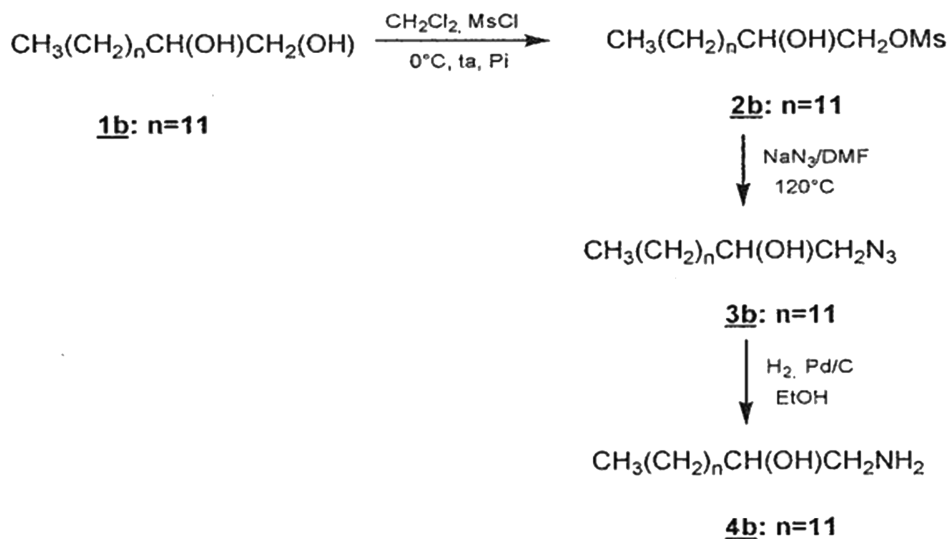


Fig. 1. Synthesis of compound 4b.

eight to twelve tissue sections from each animal. After quantifying each section of tissue, we obtained an average value representative of each animal.

2.5. Enzyme-Linked immunosorbent assay (ELISA)

The evaluation of the levels of cytokines and chemokines was carried out on the 21st dpi both in the spinal cord and in the inguinal lymph nodes of animals ($n = 7/\text{group}$). The animals' spinal cords and inguinal lymph nodes were collected and homogenized in a cytokine extraction buffer (100 mg/ml) consisted of 0.4 M NaCl (Isofar Indústria e Comércio), 0.05% Tween 20 (Isofar Indústria e Comércio), 0.5% bovine serum albumin (BSA; Sigma-Aldrich, St. Louis, MO, USA), 0.1 M phenyl-methyl-sulfonyl (PMSF; Sigma-Aldrich, St. Louis, MO, USA), 0.1 M benzethonium chloride (BC; Sigma-Aldrich, St. Louis, MO, USA), 10 mM ethylenediaminetetraacetic acid (EDTA; Sigma-Aldrich, St. Louis, MO, USA), and 20 pM aprotinin (AP; Sigma-Aldrich, St. Louis, MO, USA). Then, 3.000 g of the homogenates were centrifuged for 30 min at 4 °C and the supernatants were collected and frozen at -80°C . The concentrations of IL-17, IFN- γ , IL-10, CCL5, and CCL20 in the supernatants were quantified using the following commercial ELISA kits: for IFN- γ , IL-10, CCL20, and CCL5, we used BD Biosciences Pharmingen, (San Diego, USA), while for IL-17, eBioscience (San Diego, USA, in accordance with the manufacturer's instructions.

2.6. Inguinal lymph nodes cell count

The animals' inguinal lymph nodes were collected and weighed ($n = 7/\text{group}$). A cell suspension was prepared by macerating the inguinal lymph nodes of each animal using a Cell Strainer (Greiner bio-one, North Carolina, USA) suspended in 1 mL of PBS. The cells were counted (millions/lymph nodes) using a Neubauer camera (HBG, Germany). After counting the cells, we obtained an average value representative of each animal.

2.7. Western Blotting

On the 21st dpi, the animals' spinal cords ($n = 5/\text{group}$) were carefully dissected from the spine and stored at -80°C until use. The samples were homogenized in RIPA buffer (0.15 M NaCl, 0.05 M Tris-HCl, pH 7.2, 0.05 M EDTA, 1% Nonidet P40, 1% Triton X-100, 0.5% Triton X-100, 0.5% sodium deoxycholate, 0.1% SDS) containing protease (SigmaFAST Protease™ Inhibitor; Sigma-Aldrich, St. Louis, USA) and phosphatase inhibitors (Phosphatase Inhibitor Cocktail (100X), Cell

Signaling, Danvers, USA). The Bradford assay (Bio-Rad Laboratories Inc, Hercules, USA) was used to determine the protein concentration in the samples. For the detection of p-Akt, 30 μg of protein was submitted to SDS-PAGE for each sample, followed by electrotransfer to nitrocellulose membranes (GE Healthcare Life Sciences, Sunnyvale, USA). Blocking was performed with a solution of 4% BSA in TBST containing 0.1% Tween-20. Subsequently, the membranes were incubated overnight at 4 °C with primary anti-phospho Akt (Ser473) antibodies (1: 1000; Cell Signaling Technologies, USA) and anti-Akt (1: 1000; Cell Signaling Technologies, USA). The membranes were washed 3 times using TBST and then incubated with a secondary antibody conjugated with peroxidase, anti-rabbit (1: 2500 in TBST; Life Technologies, Carlsbad, USA). Finally, the membranes were incubated with ECL prime detection reagent (GE Healthcare Life Sciences, Sunnyvale, USA) to improve chemiluminescence. The quantification of unsaturated immunoreactive bands was performed by scanning densitometry with ImageQuant LAS 4000 (GE Healthcare Life Sciences, Sunnyvale, USA). We verified the optical densities of the detected bands using the ImageJ software (NIH, Bethesda, USA). The levels of p-Akt expression were normalized to the levels of total Akt in each sample.

2.8. Statistical analysis

The statistical software GraphPad Prism 6.0 (GraphPad Software, La Jolla, CA, USA) was used to perform the statistical analysis. Differences observed between groups for histological analysis, cytokine and chemokine levels, and Akt and pAkt expressions, were analyzed by one-way ANOVA, while the analysis of the EAE clinical score was performed by two-way ANOVA. Bonferroni's test was used in both cases as a post-hoc test. The data are presented as mean $\pm$ standard error of the mean (SEM). The level of significance was set at $p < 0.05$.

3. Results

3.1. Compound 4b inhibited the development of EAE

The clinical evaluation of EAE was performed daily in mice for 21 days. The animals were immunized with MOG₃₅₋₅₅, and those that did not receive any treatment showed onset of clinical signs of EAE, like loss of muscle tone and tail weakness, on the 10th dpi. In addition, the worsening of clinical evolution occurred from the 13th dpi, reaching the peak of the clinical score on the 17th dpi. In the group in which the animals were immunized with MOG₃₅₋₅₅ and treated with 4b, only 4 animals (30%) developed initial clinical signs of the disease, without

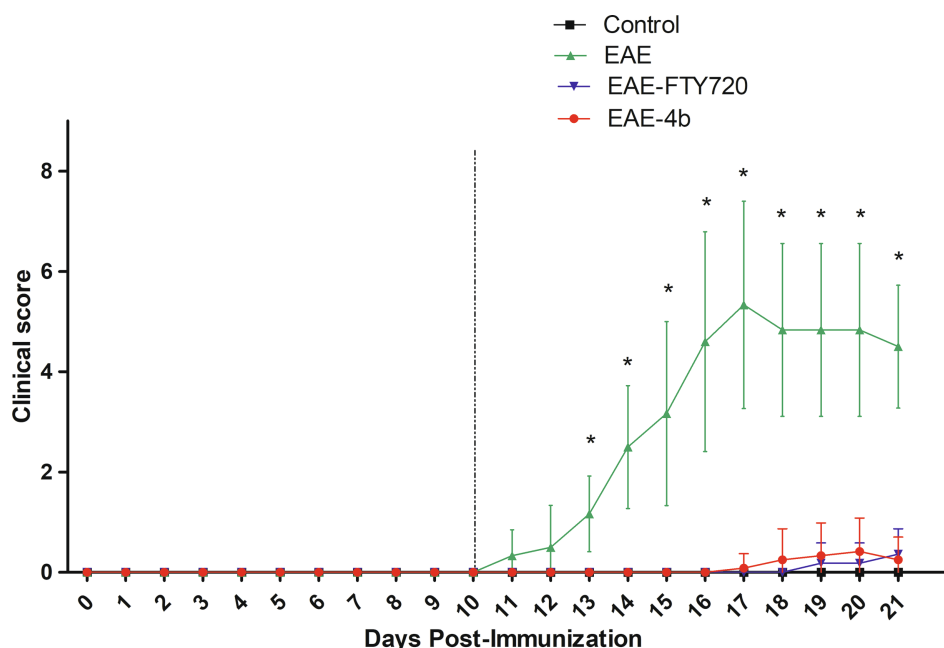


Fig. 2. Compound 4b inhibited the development of EAE. Clinical symptoms of EAE started on day 10 and peaked on day 17 after immunization.

the evolution of clinical score until the 21st dpi, similar to the observed in the FTY720 treatment ($p < 0.001$). This result shows that 4b was able to inhibit the clinical evolution of EAE (Fig. 2).

3.2. Compound 4b decreased the inflammation in the inguinal lymph nodes

Since secondary lymphoid organs are sites where immune cell activation and proliferation occurs, we analyzed the number of cells, the production of chemokines and cytokines in the homogenate, and the weight of inguinal lymph nodes in EAE animals, after treatment with 4b. Our results show that EAE animals had an increased weight of inguinal lymph nodes and a greater number of cells in these locations, compared to the control group. EAE animals showed increased production of chemokines CCL5 and CCL20 and proinflammatory cytokines IFN- γ and IL-17, compared to the controls. The EAE-4b group showed a reduction in the weight of the inguinal lymph nodes and in the cell count at these sites, similar to the control group. 4b also caused a reduction in the levels of IFN- γ and IL-17 and a raise in the levels of cytokine IL-10. Surprisingly, 4b showed a more effective action in reducing the levels of IFN- γ cytokine, although treatment with FTY720 caused suppression of immune cells in the inguinal lymph nodes of EAE-induced animals, compared to control animals (Fig. 3).

3.3. Compound 4b decreased the infiltration of the inflammatory cells and demyelination in the spinal cord

Taking into account that the migration of inflammatory cells and the demyelination process in the CNS are fundamental for the pathogenesis of MS, our next step was to evaluate the effect of compound 4b in the presence of immune cells and foci of demyelination in the spinal cord in EAE mice (Fig. 4). Through the results of H&E staining, we can see that the EAE group presented moderate cell infiltration in the parenchymal region of the spinal cord. The EAE-4b group showed discrete inflammatory infiltrates and, in some animals, those infiltrates were only visible in the perivascular areas (Fig. 4A). Expectedly, it can be observed that the compound 4b reduced the EAE inflammatory process in the spinal cord, similar to the observed in the EAE-FTY720 group (Fig. 4B; $p < 0.05$). The demyelination process in the spinal cord was observed on the 21st dpi using LFB staining. The EAE group showed foci of demyelination characterized as confluent in the perivascular regions,

while the animals of the group EAE-4b did not present foci of demyelination (Fig. 4C). Therefore, it can be observed that treatment with compound 4b reduced the destruction of myelin in the spinal cord, in a way compatible with the reduction of the inflammatory process and clinical evolution of the disease, similar to the observed in the EAE-FTY720 group ($p < 0.05$) (Fig. 4D). The histological sections of the spinal cord of the non-immunized control group were considered normal, without the presence of inflammatory foci and demyelination (Fig. 4).

3.4. Compound 4b reduced the proliferation of microglia and astrocytes in the spinal cord

It was observed gliosis in the spinal cord of animals after EAE induction, measured on the 21st dpi. The data were obtained by immunostaining astrocytes (GFAP⁺ cells) and microglia (Iba-1⁺ cells) in the spinal cord of EAE mice and control mice. The results demonstrated that 4b was able to significantly reduce the microgliosis ($p < 0.05$) (Fig. 5A) and astrocytosis ($p < 0.05$) (Fig. 5B) in the spinal cord in EAE mice, similar to that observed in the EAE-FTY720 group.

3.5. Compound 4b altered the production of chemokines and cytokines in the spinal cord in the EAE model

The levels of chemokines and of anti- and pro-inflammatory cytokines were measured in the spinal cord homogenate. The results showed that the levels of CCL5, CCL20, IFN- γ , and IL-17 increased in the spinal cord of EAE mice in comparison to the control group. As illustrated in Figs. 6, 4b was able to significantly decrease ($p < 0.05$) CCL5 and CCL20 chemokines, and IFN- γ and IL-17 cytokines in EAE mice in comparison to the EAE group that received DMSO, similar to the observed in the EAE-FTY720 group. Interestingly, in the EAE-4b group, there was less production of the cytokine IFN- γ in the spinal cord compared to the EAE-FTY720 group. IL-10 levels were also shown to increase in the spinal cord of EAE mice compared to the control group. As represented in Fig. 6 (E and J), the treatment with 4b significantly increased ($p < 0.05$) the levels of IL-10 compared to the EAE group that received DMSO in the spinal cord, similar to the observed in the EAE-FTY720 group.

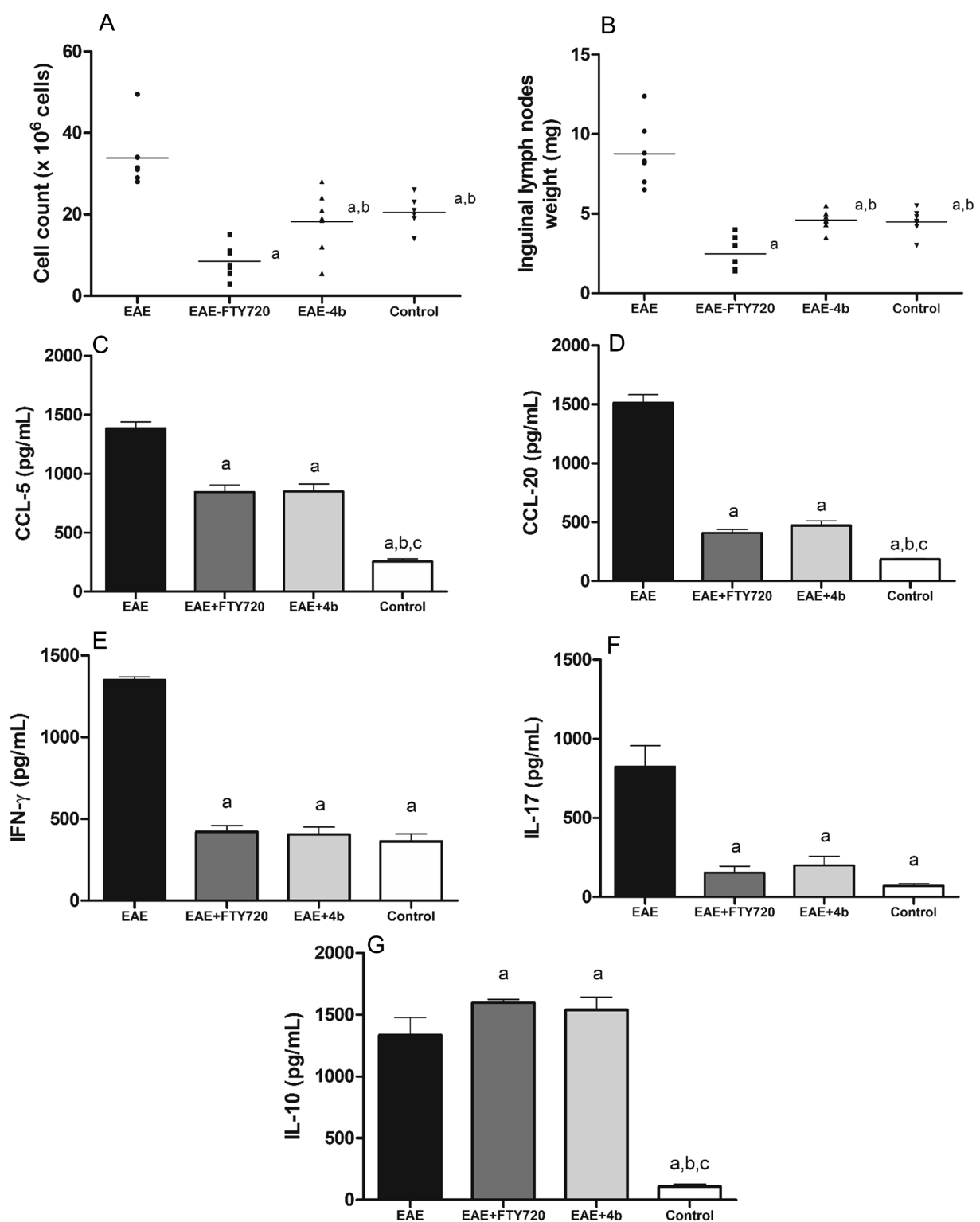


Fig. 3. Compound 4b altered the weight and cell count (A- B) and the production of chemokines and cytokines in the inguinal lymph nodes in the EAE model (n = 7/ group). 4b treatment reduces CCL5 (C) and CCL20 (D) chemokines, IFN-γ (E) and IL-17 proinflammatory (F) cytokines and enhanced IL-10 anti-inflammatory cytokine (G) levels in the inguinal lymph nodes. a = p < 0.05 versus EAE group, b = p < 0.05 versus EAE-FTY720 group, c = p < 0.05 versus EAE-4b group.

3.6. Compound 4b reduced the ratio pAkt/Akt in the spinal cord of EAE mice

Since 4b had significant effects in inhibiting the clinical evolution of EAE induced by MOG₃₅₋₅₅ in C57BL/6 mice, it was investigated whether this treatment would interfere in the phosphorylation of the Akt protein in the CNS. For this, the ratio between the levels of expression

of the phosphorylated Akt protein and the total Akt was determined in the spinal cord of EAE mice treated with FTY720 or 4b, on the 21st dpi, using the technique of Western Blotting (Fig. 7). It can be noted that the EAE group showed a higher level of pAKT expression compared to the control and treated groups with 4b, as well as FTY720 on the 21st dpi, which indicates greater activation of the PI3K/Akt pathway in EAE disease (p < 0.05). With regard to the group treated with 4b or

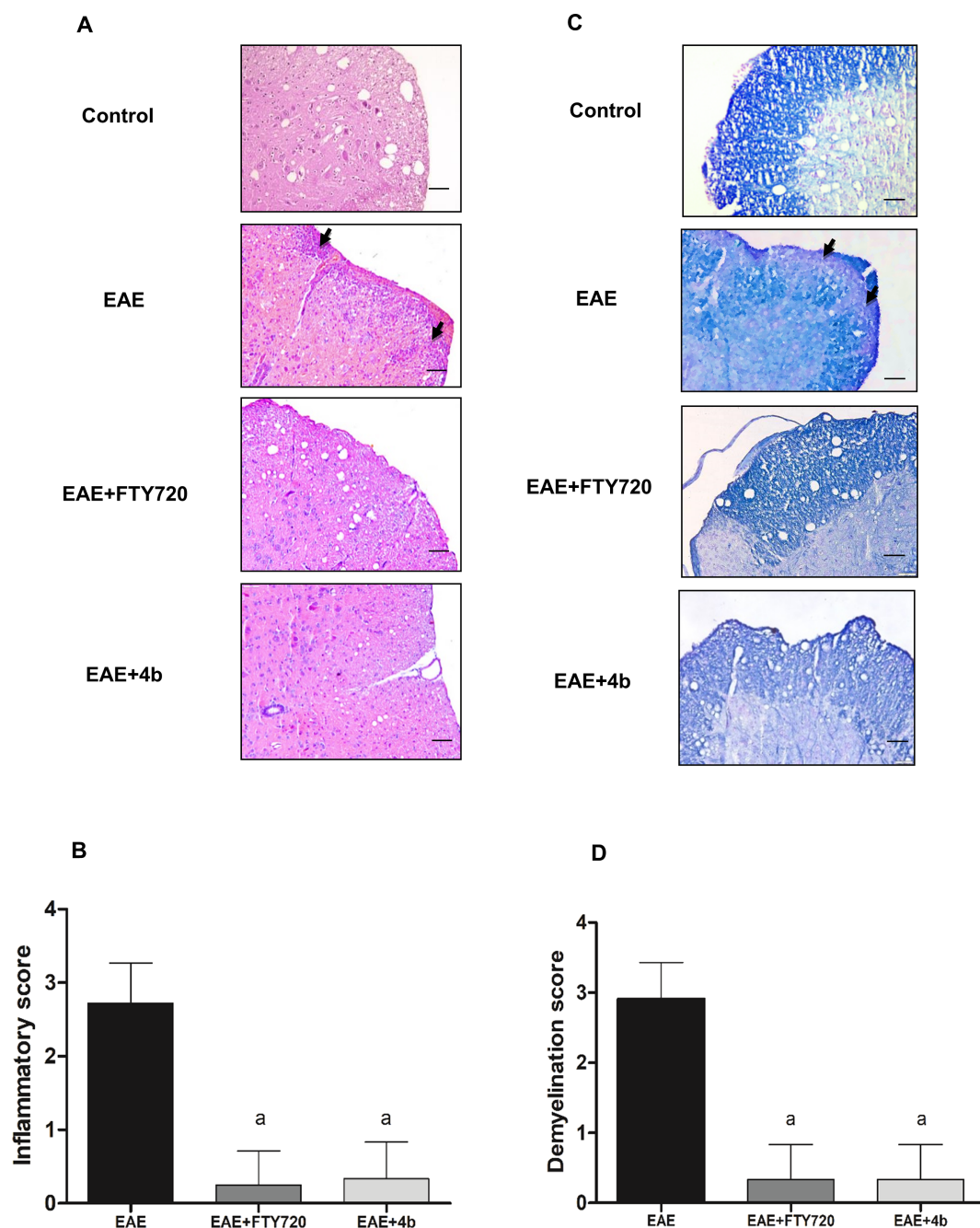


Fig. 4. Compound 4b decreased the infiltration of the inflammatory cells and demyelination in the spinal cord. 4b treatment reduces inflammatory cell infiltration (A–B) and foci of demyelination (C–D) in the spinal cord of EAE mice (n = 5/group). * Shows the significant differences in relation to EAE group (p < 0.05).

FTY720, there were no significant differences between them and the control group (Fig. 7A and B).

4. Discussion

Current treatments for MS are able to decrease relapse rates in patients who have the remitting-recurrent form, however, preventing long-term effects remains a problem; drugs for progressive forms of MS are also limited in their effectiveness. Moreover, these treatments still have a high cost for patients and public health services. Therefore, new, cheaper, and safer drugs are needed to effectively treat MS [18–20].

The compound 4b, which presents high yield and low cost, is a lipophilic amino alcohol because it has a long carbon chain. The lipophilicity generated by this condition favors a greater diffusion through biological membranes, increasing the compound's intracellular

concentration [21]. Studies have shown that lipophilic amino alcohols were able to reduce the production of inflammatory mediators such as NO and pro-inflammatory cytokines (e.g. TNF- α and IL-1 β) both in *in vitro* and *in vivo* models of inflammation [22,23]. In previous works developed in our research group, we demonstrated that lipophilic derivatives of genistein – a phytoestrogen present in soy – had different modulating effects when tested on J774.A1 macrophages strains (estimated by LPS), in which the inhibition of IL-12 and NO production probably occurred by activating of the NF- κ B pathway [24].

In our study, the treatment for 10 days with the compound 4b was able to inhibit the worsening of the clinical signs of EAE, as well as the infiltration of inflammatory immune cells and the number of demyelination foci in the CNS. The autoreactive inflammatory cells activated in the secondary lymphoid organs – such as spleen and draining lymph nodes – migrate towards the CNS, orchestrated by an increase in BBB

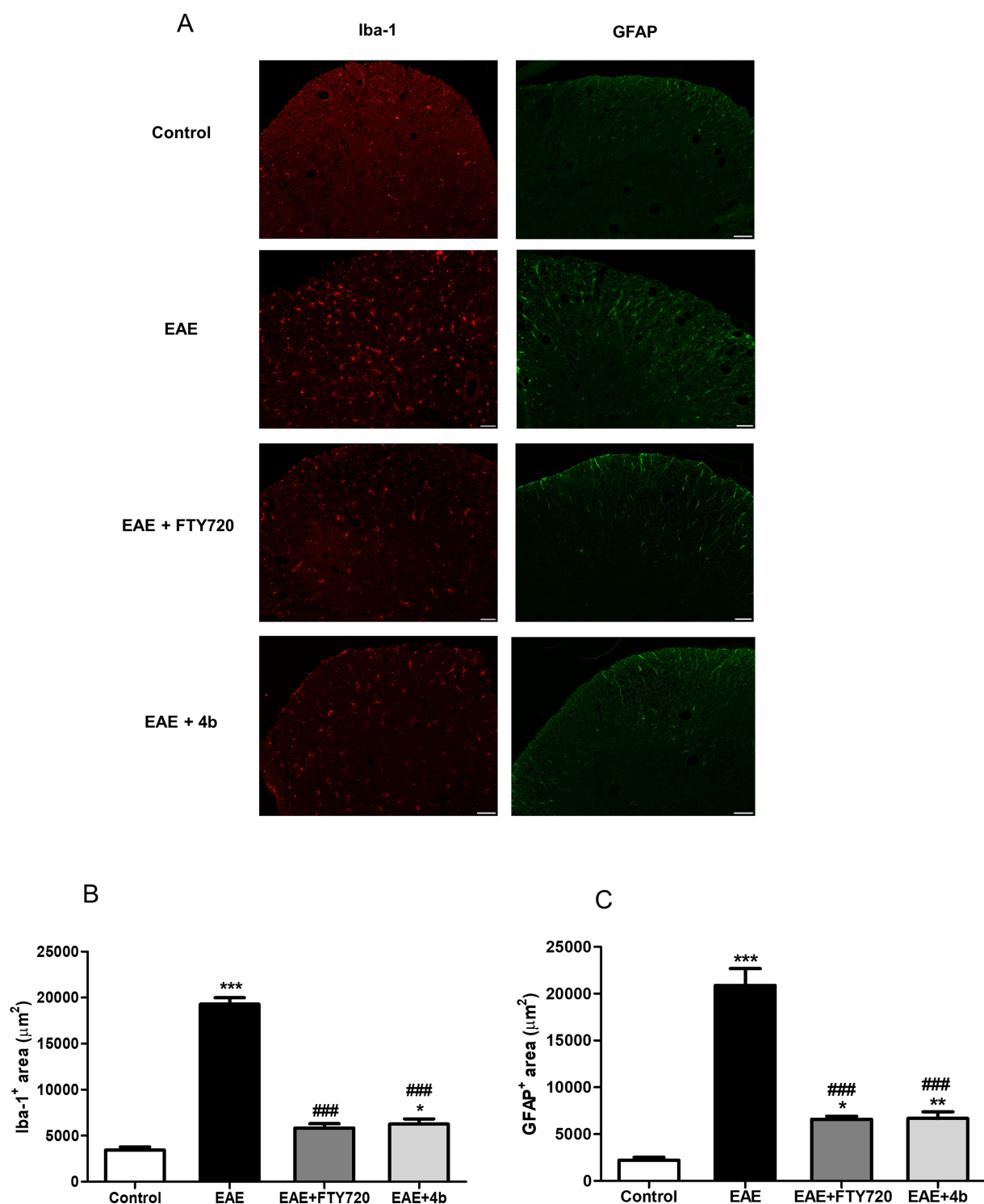


Fig. 5. Compound 4b reduced the proliferation of microglia and astrocytes in the spinal cord. (A) 4b treatment decreased astrogliosis and microgliosis. The semi-quantitative analysis ($n = 5/\text{group}$) demonstrated that 4b treatment mitigate microgliosis (B) and astrogliosis (C). * Shows the significant differences in comparison to control group ($p < 0.05$), ** ($p < 0.01$) and *** ($p < 0.001$). #Shows the significant differences in comparison to EAE group ($p < 0.05$).

permeability and chemokines production [17]. Here, we showed that the treatment with 4b promoted maintenance of the weight of the inguinal lymph nodes compared with the control group. The untreated EAE group showed an increase in the mass of inguinal lymph nodes, which suggests a greater presence of immune cells in the lymph nodes of this group compared to the EAE-4b and control groups. We believe that the treatment with compound 4b for 10 days from the onset of the clinical signs of EAE may have influenced the outflow of T lymphocytes

from the thymus to the periphery. Studies had already demonstrated a regulatory effect of the treatment with the FTY720 compound in reducing the egress of lymphocytes from the thymus to the periphery in an animal model of skin graft rejection [25]. Consequently, the reduction in the activation process of lymphocytes in the periphery would be observed due to the egress of these cells from the thymus and the reduced presentation of antigens that participate in the induction of the EAE model in secondary lymphoid organs. Considering these data, the

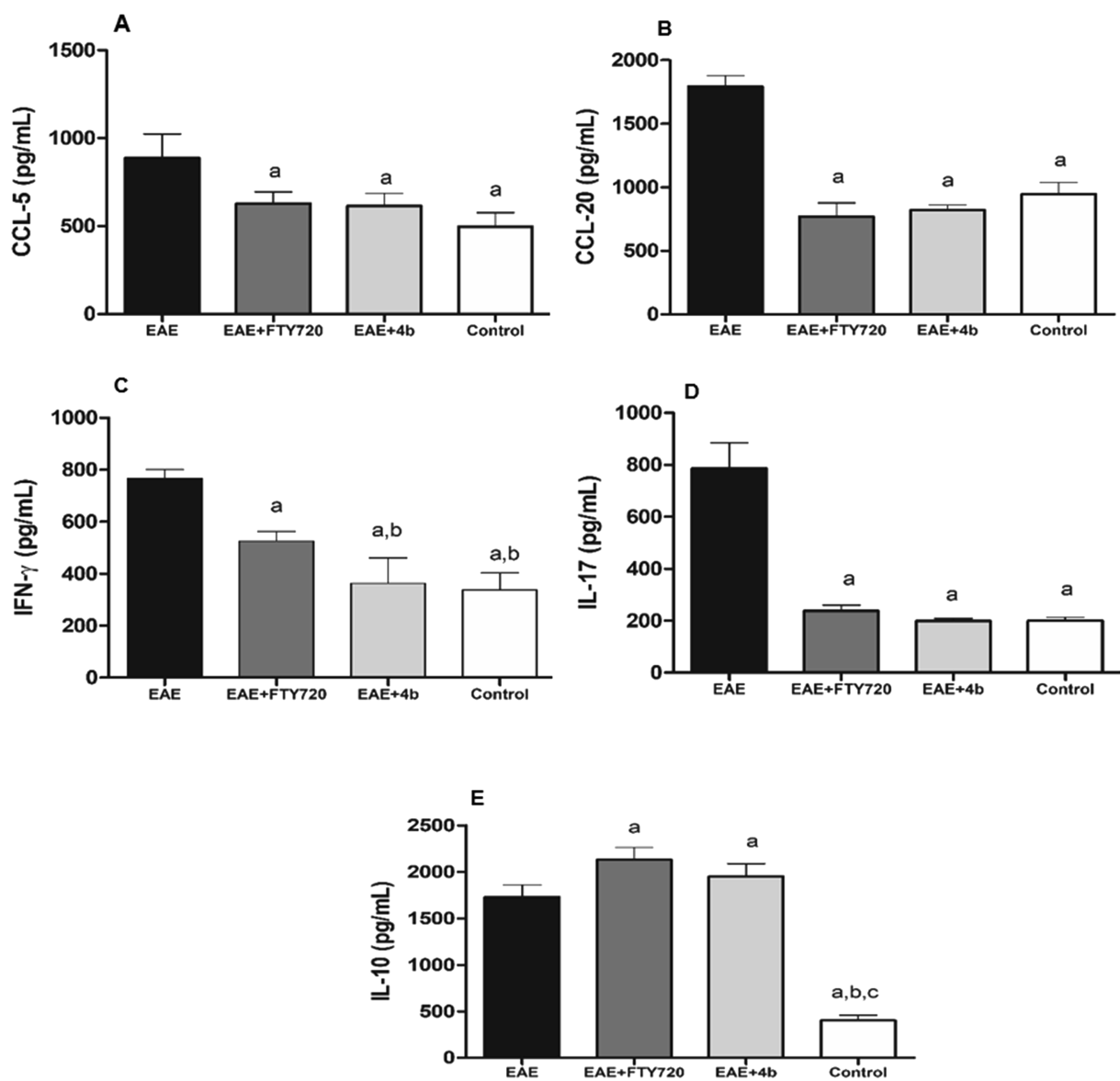


Fig. 6. Compound 4b altered the production of chemokines and cytokines in the spinal cord in the EAE model ($n = 7/\text{group}$). 4b treatment reduces CCL5 (A) and CCL20 (B) chemokines, IFN- γ (C) and IL-17 pro-inflammatory (D) cytokines and enhanced IL-10 anti-inflammatory cytokine (E) levels in the spinal cords. $a = p < 0.05$ versus EAE group, $b = p < 0.05$ versus EAE-FTY720 group, $c = p < 0.05$ versus EAE-4b group.

start of the treatment with 4b on the day of the onset of signs of EAE was able to inhibit the assembly of an inflammatory response in the peripheral lymphoid organs. This data is compatible with the higher production of pro-inflammatory cytokines (IFN- γ and IL-17) in the inguinal lymph nodes and with the lower production of IL-10 in the EAE group if compared to the EAE-4b group. We signal the need for a new study based on this work to investigate the action of compound 4b in the egress of lymphocytes from the thymus and in their appearance and clonal expansion in the periphery [25]. It was reported that receptors of the chemokines found in immune cells are involved in a variety of inflammatory diseases of the CNS and contribute to the pathogenesis of EAE and other autoimmune disorders, such as systemic lupus erythematosus [26]. The Maraviroc compound, a CCL5 chemokine receptor (CCR5) antagonist (RANTES), improved the MOG-induced EAE by inhibiting the migration of inflammatory cells to the spinal cord [27]. Here, we demonstrated that the compound 4b significantly reduced the production of chemokines CCL5 and CCL20, which have been shown to be important cell chemoattractant factors in the EAE model [28,29]. Microglia and astrocytes are important glial cells for immune

responses and protection of the CNS [30]. However, persistently activated microglia/macrophages produce toxic substances that are harmful to the CNS [31]. Astrocytes are the most abundant cell population in the CNS. The appearance of lesions in the CNS can cause astrocytic activation, which can be identified by the increased expression of the glial fibrillar acid protein (GFAP) [32]. Microglial activation, together with astrocyte activation, can act in the polarization of the infiltrating T CD4+ cells in the CNS for pro-inflammatory phenotypes and/or strengthen their effector effect [33–35]. In addition, soluble mediators produced and released by astrocytes can target T CD8+ and B cell responses, including cytotoxicity, maturation, and survival [36–38]. In our study, compound 4b reduced the expression of Iba-1 (microglia) and GFAP (astrocytes) markers in the spinal cord of EAE mice. The proinflammatory cytokines IFN- γ and IL-17, already well characterized as protagonists of EAE [39], were reduced in the spinal cords and in the inguinal lymph nodes after 10 days of treatment with 4b in the EAE group.

A study showed that C57Bl/6 *knockout* mice for CCR5 presented an inhibition of the development of EAE, with reduced infiltration of

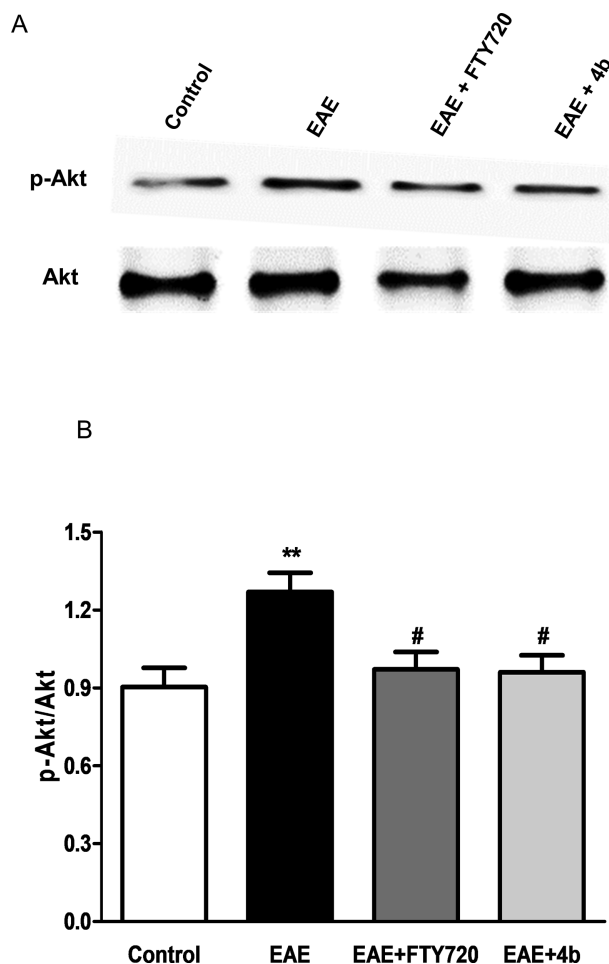


Fig. 7. Compound 4b reduced the ratio pAkt/Akt in the spinal cord of EAE mice. (A) Representative western blot showing expression of the indicated proteins in different groups (n = 5/groups). (B) Quantitative densitometric analysis of normalized phospho-Akt with respect to the total Akt. ** Shows the significant differences in comparison to control group (p < 0.01). #Shows the significant differences in comparison to EAE group (p < 0.05).

immune cells in the CNS, preventing glial cell activation and the production of proinflammatory mediators [40].

Surprisingly, we observed a better action of the compound 4b in reducing the production of the cytokine IFN- γ in the spinal cord and inguinal lymph nodes in comparison to FTY720, a drug that has already been established as a treatment for MS. Daniel and colleagues observed a negative regulation in the secretion of the cytokine IL-12p70 – essential for the production of cytokines of the Th1 profile like IFN- γ – in a Th1-mediated 2,4,6-trinitrobenzene sulfonic acid (TNBS) colitis model after treatment with the compound FTY720 [41]. In another study using an autoimmune myasthenia gravis model, FTY720 also reduced cytokines levels of IL-2, IFN- γ , and IL-6 in tissue and serum [42]. Thus, we point out that the greater lipophilia of the analog may have been responsible for its greater action in the production of inflammatory mediators [43]. MS and EAE are chronic inflammatory autoimmune diseases that lead to the process of axonal demyelination, causing losses in the nerve conduction responsible for the symptoms of diseases, such as physical and neurological disabilities. The severity of such symptoms is related to the number and extent of demyelination plaques in the CNS of patients. Thus, the analysis of the CNS demyelination process is a strong indication of the neurodegeneration promoted by the inflammatory condition. Previous studies by our group have evaluated the CNS demyelination process in EAE animals after treatment with mitoxantrone [29] and genistein [17], concluding that

they were able to reduce the outbreaks of demyelination leading to greater neuronal protection and, consequently, to a reduction in the worsening of the disease. In this work, we observed that compound 4b was reduced the demyelination process in the spinal cord of EAE animals, which was compatible with the lesser symptomatology observed by monitoring the animals' clinical signs after EAE induction. Based on the association of such data, we showed that compound 4b protected the neuronal functions of animals in which the EAE model was induced. The cytokine IL-10, which acts as an anti-inflammatory agent, was elevated in the inguinal lymph nodes and spinal cords after treatment with compound 4b. It is already known that the elevation of this cytokine during inflammatory reactions developed in EAE may be related to the suppression of inflammatory cascades and demyelination process, culminating in the clinical recovery of the animals [27,39]. During the inflammatory process, cytokines and growth factors can activate T-cell signaling pathways. The PI3K/Akt/mTOR signaling pathway comprises a set of intracellular kinases capable of stimulating cellular responses to conditions in the extracellular environment. In mammalian cells, phosphorylation of the PI3K protein leads to the production of PIP3 (phosphatidylinositol 3,4,5-triphosphate). Elevation of PIP3 leads to the recruitment of 3-phosphoinositide-dependent protein kinase 1 (PDK1) and Akt protein to the cell membrane and subsequent phosphorylation of Akt by PDK1. Once phosphorylated, Akt acts in the phosphorylation process of the heteromeric protein complex, commonly referred to as mTOR complex 1 (mTORC1) and mTOR complex 2 (mTORC2), involved in the control of a wide variety of cellular processes, regulating metabolism, cell growth, cell differentiation, and cell survival [44–46]. This signaling pathway also plays an important role in regulating the proliferation and migration of T cells in inflammatory conditions [47]. Recent studies have also found that PI3K/Akt/mTOR positively regulates Th17 differentiation and the IL-17 cytokine levels in vivo and in vitro via multiple mechanisms [48]. Moreover, it has been shown that compounds capable of inhibiting PI3K and/or mTOR complex in TCD4⁺ cells increased the polarization of T naive to T regulatory cells and increased the production of anti-inflammatory cytokines, like IL-10. In contrast, the activation of PI3K and/or mTOR enhances the differentiation and effector functions of pro-inflammatory cells, as Th1 and Th17 cells [49–51]. mTOR inhibitors have shown good efficacy in the treatment of inflammatory conditions, such as graft rejection in various forms of transplants, renal dysfunction, atherosclerosis, autoimmune diseases, and other disorders of neurological origin, such as epilepsy, Alzheimer's disease, Parkinson's disease, among others [44]. Here, we found that treatment with the new lipophilic amino alcohol 4b suppressed the Akt protein activation, reducing the levels of its phosphorylated form (pAkt) in the spinal cord of animals with EAE, similar to the treatment with FTY720. Our results also indicate a regulatory action of 4b in the PI3K/Akt/mTOR pathway in inflammatory sites, controlling the production of inflammatory cytokines and reducing the worsening of EAE and, possibly, MS.

5. Conclusion

Our results showed that treatment with the new lipophilic amino alcohol 4b, started from the onset of clinical signs of EAE, was able to inhibit the worsening of the clinical signs of the disease through the reduction in the levels of chemokines and pro-inflammatory cytokines, as well as the reduction of microgliosis and, astrogliosis. Moreover, 4b promoted an increase in the production of the anti-inflammatory cytokine IL-10. These results can be related by the modulation of the PI3K/Akt axis in the spinal cords of EAE animals. Therefore, this work presented promising data for the compound 4b in the treatment of yet another inflammatory disease, the EAE, the main model used for the study of MS.

Luan Cristian Silva: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Data curation, Writing - original draft, Visualization. **Isabel Vieira Assis Lima:** Methodology, Validation, Formal analysis, Investigation, Writing - original draft. **Maria Carolina Machado Silva:** Formal analysis. **Taís Arthur Corrêa:** Resources. **Viviane Passos Souza:** Formal analysis. **Mauro Vieira Almeida:** Resources. **Antônio Carlos Pinheiro Oliveira:** Methodology, Validation, Resources, Writing - review & editing, Supervision, Funding acquisition. **Ana Paula Ferreira:** Conceptualization, Methodology, Validation, Resources, Writing - review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.intimp.2020.106919>.

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